

A Theoretical and Numerical Study of the Su–Schrieffer–Heeger Model

Summer Research Internship Report

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1 Introduction

The Su–Schrieffer–Heeger (SSH) model is a one-dimensional tight-binding model that serves as one of the simplest theoretical frameworks for studying topological phases of matter. Originally introduced to describe the electronic properties of polyacetylene [1], the model has since become one of the canonical examples for understanding topological band theory and one-dimensional topological insulators [2]. Despite its simple structure, the SSH model exhibits non-trivial topological behaviour and provides an accessible platform for understanding concepts such as energy bands, topological phase transitions, and edge states, which are central topics in the study of topological insulators [3, 4]. This report presents the derivation and numerical investigation of the SSH model starting from its real-space tight-binding Hamiltonian. The Hamiltonian is expressed in matrix form for a finite lattice and subsequently reformulated using the tensor-product representation of the unit-cell and sublattice degrees of freedom. By imposing periodic boundary conditions and applying Bloch’s theorem, the bulk momentum-space Hamiltonian is obtained, from which the energy dispersion relation is derived. Finally, the Hamiltonian is expressed in terms of Pauli matrices, leading to the vector representation that forms the basis for describing the topological properties of the model [2].

In this internship, the SSH model is studied from both analytical and numerical perspectives. First, the bulk momentum-space Hamiltonian, energy dispersion, chiral symmetry, winding number, and bulk-boundary correspondence are derived from the real-space tight-binding Hamiltonian. Numerical diagonalization of finite SSH chains is then performed to investigate the energy spectrum and edge-state localization. Finally, the SSH model is extended by introducing a staggered on-site potential, and its effects on the energy spectrum, eigenstates, chiral symmetry, and topological properties are examined.

2 The Su–Schrieffer–Heeger Model

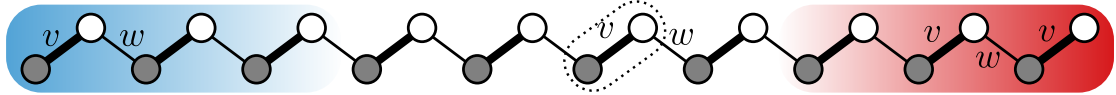


Figure 1: Geometry of the Su–Schrieffer–Heeger (SSH) model. Each unit cell consists of two sublattice sites, A (grey) and B (white), with alternating intracell (v) and intercell (w) hopping amplitudes. Reproduced from Asbóth *et al* [2].

2.1 Real-Space Tight-Binding Hamiltonian

The Su–Schrieffer–Heeger (SSH) model describes a one-dimensional lattice composed of N unit cells, with each unit cell containing two inequivalent lattice sites belonging to sublattices A and B, as shown in Fig. 1. The model assumes that a particle can hop only between nearest-neighbour sites. Due to the alternating bond strengths, two distinct hopping amplitudes are introduced: the intracell hopping amplitude v , which couples the A and B sites within the same unit cell, and the intercell hopping amplitude w , which couples the B site of one unit cell to the A site of the neighbouring unit cell. [1, 2] Within the tight-binding approximation [2, 5], the basis states are denoted by $|m, A\rangle$ and $|m, B\rangle$, representing a particle localized on the A or B site of the m^{th}

unit cell, respectively. Neglecting the on-site potentials and restricting the hopping to nearest neighbours, the real-space Hamiltonian of the SSH model is written as

$$\hat{H} = v \sum_{m=1}^N (|m, B\rangle\langle m, A| + \text{h.c.}) + w \sum_{m=1}^{N-1} (|m+1, A\rangle\langle m, B| + \text{h.c.}) \quad (1)$$

where h.c. represents the Hermitian conjugate.

2.2 Matrix Representation of the Hamiltonian

To obtain an explicit matrix representation of the SSH Hamiltonian, we consider a finite chain and express the Hamiltonian in a localized basis. For a chain consisting of N unit cells, the Hilbert space has dimension of $2N$, since each unit cell contains two sublattice sites. As an example, we first consider a chain with $N = 4$ unit cells. The basis states are chosen as $\{|1, A\rangle, |1, B\rangle, |2, A\rangle, |2, B\rangle, |3, A\rangle, |3, B\rangle, |4, A\rangle, |4, B\rangle\}$. Using Eq. (1), the action of the Hamiltonian on each basis state is obtained as

$$\begin{aligned} \hat{H}|1, A\rangle &= v|1, B\rangle, \\ \hat{H}|1, B\rangle &= v|1, A\rangle + w|2, A\rangle, \\ \hat{H}|2, A\rangle &= v|2, B\rangle + w|1, B\rangle, \\ \hat{H}|2, B\rangle &= v|2, A\rangle + w|3, A\rangle, \\ \hat{H}|3, A\rangle &= v|3, B\rangle + w|2, B\rangle, \\ \hat{H}|3, B\rangle &= v|3, A\rangle + w|4, A\rangle, \\ \hat{H}|4, A\rangle &= v|4, B\rangle + w|3, B\rangle, \\ \hat{H}|4, B\rangle &= v|4, A\rangle \end{aligned}$$

Each equation gives one column of the Hamiltonian matrix in the chosen basis. For example, $\hat{H}|1, A\rangle = v|1, B\rangle$ means that the first column has only one nonzero element, $H_{21} = v$, while $\hat{H}|1, B\rangle = v|1, A\rangle + w|2, A\rangle$ gives the second column $H_{12} = v$ and $H_{32} = w$. Repeating this procedure for all basis states gives

$$H = \begin{pmatrix} 0 & v & 0 & 0 & 0 & 0 & 0 & 0 \\ v & 0 & w & 0 & 0 & 0 & 0 & 0 \\ 0 & w & 0 & v & 0 & 0 & 0 & 0 \\ 0 & 0 & v & 0 & w & 0 & 0 & 0 \\ 0 & 0 & 0 & w & 0 & v & 0 & 0 \\ 0 & 0 & 0 & 0 & v & 0 & w & 0 \\ 0 & 0 & 0 & 0 & 0 & w & 0 & v \\ 0 & 0 & 0 & 0 & 0 & 0 & v & 0 \end{pmatrix}$$

The same procedure can be extended to an arbitrary number of unit cells, yielding the $2N \times 2N$ tri-diagonal Hamiltonian of the SSH model.

2.3 Tensor Product Representation

The matrix representation derived in the previous section becomes increasingly difficult as the number of unit cells increases. A more compact formulation is obtained by separating the external

degrees of freedom, corresponding to the unit-cell index, from the internal degrees of freedom, corresponding to the two sublattice sites. The localized basis states may therefore be written as [2]

$$|m, \alpha\rangle = |m\rangle \otimes |\alpha\rangle, \quad \alpha \in \{A, B\}$$

where $|m\rangle$ denotes the state associated with the m^{th} unit cell and $|\alpha\rangle$ represents the internal degree of freedom.

The Pauli matrices are defined as

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Using this decomposition, the intracell hopping operator may be written as

$$\begin{aligned} |m, B\rangle\langle m, A| &= (|m\rangle \otimes |B\rangle)(\langle m| \otimes \langle A|) \\ &= (|m\rangle\langle m|) \otimes (|B\rangle\langle A|) \end{aligned}$$

where the tensor-product identity $(A \otimes B)(C \otimes D) = AC \otimes BD$ has been used. Similarly, the intercell hopping operator becomes

$$\begin{aligned} |m+1, A\rangle\langle m, B| &= (|m+1\rangle \otimes |A\rangle)(\langle m| \otimes \langle B|) \\ &= (|m+1\rangle\langle m|) \otimes (|A\rangle\langle B|). \end{aligned}$$

Assuming the internal basis is

$$|A\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |B\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Evaluating the outer products explicitly yields:

$$|B\rangle\langle A| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} \frac{\sigma_x + i\sigma_y}{2} &= \frac{1}{2} \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right) \\ &= \frac{1}{2} \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right) \\ &= \frac{1}{2} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

$$\frac{\sigma_x + i\sigma_y}{2} = |A\rangle\langle B|$$

Similarly we can show that

$$\frac{\sigma_x - i\sigma_y}{2} = |B\rangle\langle A|$$

Substituting these relations into the real-space Hamiltonian,

$$\begin{aligned}
\hat{H} &= v \sum_{m=1}^N |m, B\rangle\langle m, A| + w \sum_{m=1}^{N-1} |m+1, A\rangle\langle m, B| + \text{h.c.} \\
&= v \sum_{m=1}^N |m\rangle\langle m| \otimes |B\rangle\langle A| + w \sum_{m=1}^{N-1} |m+1\rangle\langle m| \otimes |A\rangle\langle B| + \text{h.c.} \\
&= v \sum_{m=1}^N |m\rangle\langle m| \otimes \frac{\sigma_x - i\sigma_y}{2} + w \sum_{m=1}^{N-1} |m+1\rangle\langle m| \otimes \frac{\sigma_x + i\sigma_y}{2} + \text{h.c.}
\end{aligned}$$

Including the Hermitian-conjugate terms gives

$$\hat{H} = v \sum_{m=1}^N |m\rangle\langle m| \otimes \sigma_x + w \sum_{m=1}^{N-1} \left(|m+1\rangle\langle m| \otimes \frac{\sigma_x + i\sigma_y}{2} + \text{h.c.} \right) \quad (2)$$

This tensor-product representation clearly separates the external (unit-cell) and internal (sublattice) degrees of freedom and provides a convenient starting point for deriving the bulk momentum-space Hamiltonian.

2.4 Bulk Hamiltonian

The real-space Hamiltonian derived in the previous section describes a finite chain with open boundaries. To study the properties of the bulk independently of the edges, periodic boundary conditions are imposed by identifying the first and last unit cells[2, 5]. This restores translational symmetry to the lattice, allowing the application of Bloch’s theorem. Under periodic boundary conditions, also known as Born–von Kármán boundary conditions, the hopping between the last and first unit cells is included, and the bulk Hamiltonian is written as

$$\hat{H}_{bulk} = \sum_{m=1}^N (v|m, B\rangle\langle m, A| + w|(m \bmod N) + 1, A\rangle\langle m, B| + \text{h.c.}) \quad (3)$$

Unlike the open-chain Hamiltonian, Eq. (3) possesses discrete translational symmetry, making the crystal momentum a good quantum number. Consequently, the eigenstates of the system may be expressed in terms of Bloch states, allowing the Hamiltonian to be transformed into momentum space.

2.5 Momentum-Space Hamiltonian

Since the bulk Hamiltonian possesses translational symmetry, Bloch’s theorem can be applied [2, 5]. The external degree of freedom is transformed from the real-space basis to the momentum-space basis through the discrete Fourier transform,

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{m=1}^N e^{ikm} |m\rangle, \quad k = \frac{2\pi n}{N}, \quad n = 0, 1, \dots, N-1 \quad (4)$$

The eigenstates of the bulk Hamiltonian can therefore be written as

$$|\Psi_n(k)\rangle = |k\rangle \otimes |u_n(k)\rangle; \quad |u_n(k)\rangle = a_n(k)|A\rangle + b_n(k)|B\rangle \quad (5)$$

where the spinor $|u_n(k)\rangle$ describes the internal sublattice degree of freedom. The momentum-space Hamiltonian is obtained by projecting the bulk Hamiltonian onto the Bloch basis,

$$\hat{H} = \langle k | \hat{H}_{\text{bulk}} | k \rangle$$

Substituting Eq. (3) into the above expression and evaluating the Fourier sums,

$$\langle k | = \frac{1}{\sqrt{N}} \sum_{m=1}^N e^{-ikm} \langle m |$$

For the operator,

$$\sum_{m=1}^N |m\rangle \langle m|$$

we have,

$$\begin{aligned} \langle k | \left(\sum_{m=1}^N |m\rangle \langle m| \right) | k \rangle &= \frac{1}{N} \sum_{n,m,l} e^{-ikn} e^{ikl} \langle n | m \rangle \langle m | l \rangle \\ &= \frac{1}{N} \sum_m e^{-ikm} e^{ikm} \\ &= \frac{1}{N} \sum_m 1 \\ &= 1 \end{aligned}$$

Now evaluate,

$$\sum_m |m+1\rangle \langle m|$$

Using the Bloch basis,

$$\begin{aligned} \langle k | \left(\sum_{m=1}^N |m+1\rangle \langle m| \right) | k \rangle &= \frac{1}{N} \sum_{n,m,l} e^{-ikn} e^{ikl} \langle n | m+1 \rangle \langle m | l \rangle \\ &= \frac{1}{N} \sum_m e^{-ik(m+1)} e^{ikm} \\ &= \frac{1}{N} \sum_m e^{-ik} \\ &= e^{-ik} \end{aligned}$$

Similarly,

$$\langle k | \left(\sum_{m=1}^N |m+1\rangle \langle m| \right) | k \rangle = e^{ik}$$

Now substituting these results into the Hamiltonian gives,

$$\begin{aligned} H(k) &= v (|B\rangle \langle A| + |A\rangle \langle B|) \\ &\quad + w (e^{-ik} |A\rangle \langle B| + e^{ik} |B\rangle \langle A|) \end{aligned}$$

Using $|A\rangle\langle B| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $|B\rangle\langle A| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ the Hamiltonian becomes,

$$H(k) = \begin{pmatrix} 0 & v + we^{-ik} \\ v + we^{ik} & 0 \end{pmatrix} \quad (6)$$

the Schrödinger equation in momentum space is therefore

$$H|u_n(k)\rangle = E(k)|u_n(k)\rangle$$

This determines the bulk energy spectrum and the corresponding eigenstates of the SSH model.

2.6 Dispersion Relation

The energy spectrum of the SSH model is obtained by solving the momentum-space Schrödinger equation

$$H(k)|u(k)\rangle = E(k)|u(k)\rangle, \quad (7)$$

where the momentum-space Hamiltonian is given by

$$H(k) = \begin{pmatrix} 0 & v + we^{-ik} \\ v + we^{ik} & 0 \end{pmatrix}. \quad (8)$$

The corresponding energy eigenvalues are obtained from the characteristic equation

$$\det(H(k) - EI) = 0, \quad (9)$$

which yields

$$\begin{vmatrix} -E & v + we^{-ik} \\ v + we^{ik} & -E \end{vmatrix} = 0, \\ E^2 - (v + we^{-ik})(v + we^{ik}) = 0.$$

Expanding the product,

$$\begin{aligned} (v + we^{-ik})(v + we^{ik}) &= v^2 + vw(e^{ik} + e^{-ik}) + w^2 \\ &= v^2 + w^2 + 2vw \cos k. \end{aligned}$$

Hence,

$$E^2(k) = v^2 + w^2 + 2vw \cos k, \quad (10)$$

and the dispersion relation is

$$E_{\pm}(k) = \pm \sqrt{v^2 + w^2 + 2vw \cos k}. \quad (11)$$

The positive and negative branches correspond to the conduction and valence bands, respectively. The minimum separation between these two bands occurs at the Brillouin-zone boundary, $k = \pm\pi$, giving the bulk energy gap

$$\Delta = 2|v - w|. \quad (12)$$

Eq. (12) shows that the energy gap decreases continuously as the hopping amplitudes approach one another and vanishes at

$$v = w.$$

The closing of the bulk band gap marks the critical point separating two distinct topological phases of the SSH model. The dispersion relation given by Eq. (11) is shown in Fig. 2 for five different choices of the hopping amplitudes. For $v > w$, the system is in the trivial insulating phase, with a finite bulk energy gap. As v approaches w , the gap closes at $k = \pm\pi$, signalling the topological phase transition. When $w > v$, the gap reopens, indicating the transition into the topological phase.

2.7 Vector representation of the Hamiltonian

The bulk Hamiltonian obtained in Eq. (3) may be expressed in terms of the Pauli matrices. Using Euler’s identities

$$e^{\pm ik} = \cos k \pm i \sin k,$$

the momentum-space Hamiltonian becomes

$$\begin{aligned} H(k) &= \begin{pmatrix} 0 & v + w(\cos k - i \sin k) \\ v + w(\cos k + i \sin k) & 0 \end{pmatrix} \\ &= (v + w \cos k) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + w \sin k \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \end{aligned}$$

Recognizing the matrices as the Pauli matrices σ_x and σ_y ,

$$H(k) = (v + w \cos k) \sigma_x + w \sin k \sigma_y. \quad (13)$$

The Hamiltonian may therefore be written in the compact form [2]

$$H(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}, \quad (14)$$

where

$$\mathbf{d}(k) = (d_x(k), d_y(k), d_z(k))$$

and

$$\begin{aligned} d_x(k) &= v + w \cos k, \\ d_y(k) &= w \sin k, \\ d_z(k) &= 0. \end{aligned}$$

As the crystal momentum k varies across the first Brillouin zone, the vector $\mathbf{d}(k)$ traces a closed trajectory in the d_x - d_y plane [2]. The topology of this trajectory forms the basis for distinguishing the trivial and topological phases of the SSH model. Specifically, the winding of the curve about the origin determines the topological invariant of the system, which is discussed in the following section.

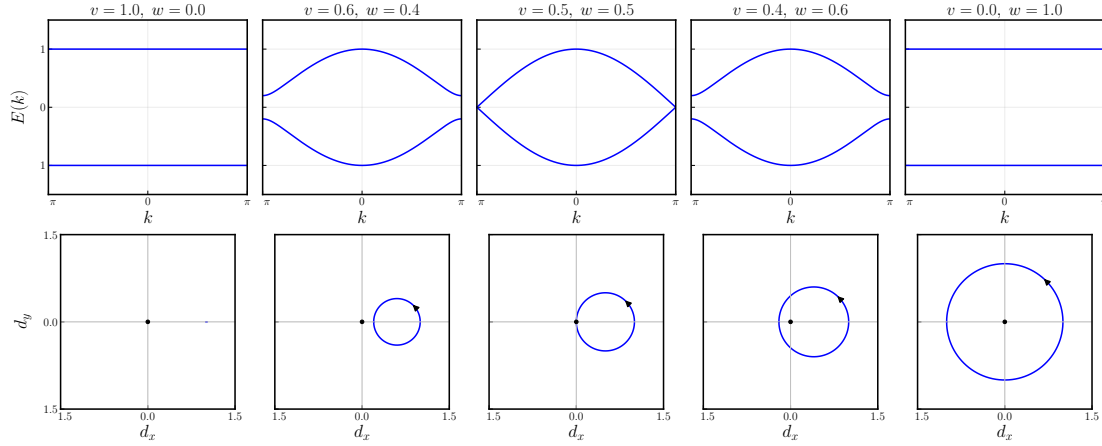


Figure 2: Energy dispersion (top row) and corresponding winding trajectories of $\mathbf{d}(k)$ in the (d_x, d_y) plane (bottom row) for the SSH model with different intracell v and intercell w hopping amplitudes. The topology of the winding trajectory changes as the relative hopping amplitudes are varied. At the critical point $v = w$, the bulk energy gap closes and the trajectory passes through the origin, marking the topological phase transition.

2.8 Chiral Symmetry

In addition to translational symmetry, the SSH model possesses an important internal symmetry known as *chiral symmetry* [2]. This symmetry plays a central role in determining the topological properties of the model and is responsible for the existence and protection of zero-energy edge states. Unlike translational symmetry, which is associated with the periodicity of the lattice, chiral symmetry relates the two sublattices of the SSH chain and imposes a specific structure on the Hamiltonian.

Unlike ordinary unitary symmetries, which satisfy

$$\hat{U}\hat{H}\hat{U}^\dagger = \hat{H},$$

chiral symmetry is characterized by an additional minus sign,

$$\hat{\Gamma}\hat{H}\hat{\Gamma}^\dagger = -\hat{H}.$$

This distinction has profound consequences for the spectrum of the Hamiltonian. While ordinary symmetries partition the Hilbert space into independent symmetry sectors, chiral symmetry instead relates eigenstates with positive and negative energies, leading to a spectrum that is symmetric about zero energy.

Definition: A Hamiltonian is said to possess chiral symmetry if there exists a unitary operator $\hat{\Gamma}$ satisfying

$$\hat{\Gamma}\hat{H}\hat{\Gamma}^\dagger = -\hat{H}. \quad (15)$$

Since the chiral operator is unitary,

$$\hat{\Gamma}^\dagger\hat{\Gamma} = \hat{\Gamma}\hat{\Gamma}^\dagger = I,$$

Eq. (15) may be written equivalently as

$$\{\hat{\Gamma}, \hat{H}\} = 0$$

where

$$\{A, B\} = AB + BA$$

denotes the anticommutator.

Conditions: For the SSH model, the chiral operator satisfies the following properties.

1. The operator $\hat{\Gamma}$ is unitary and Hermitian,

$$\hat{\Gamma}^\dagger = \hat{\Gamma}, \quad \hat{\Gamma}^2 = I.$$

The requirement that the chiral operator be both unitary and Hermitian ensures that $\Gamma^2 = I$. If this condition were not satisfied, the operator Γ^2 would itself generate an ordinary unitary symmetry of the Hamiltonian. Thus, imposing $\Gamma^2 = I$ guarantees that chiral symmetry represents a distinct symmetry class rather than an ordinary unitary symmetry.

2. The operator acts locally within each unit cell and distinguishes the two sublattices through its eigenvalues. Since the chiral operator is Hermitian and satisfies

$$\Gamma^2 = I,$$

its eigenvalues are restricted to ± 1 . Choosing the sublattice basis to coincide with the eigenbasis of the chiral operator, the basis states satisfy

$$\Gamma|A\rangle = |A\rangle, \quad \Gamma|B\rangle = -|B\rangle.$$

Thus, the A and B sublattices correspond to the positive- and negative-chirality eigenspaces, respectively. In this basis the chiral operator is represented by

$$\Gamma = \sigma_z.$$

To verify that $\Gamma = \sigma_z$ indeed generates chiral symmetry, consider the momentum-space Hamiltonian of the SSH model,

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y. \quad (16)$$

Since σ_z is Hermitian and unitary,

$$\sigma_z^\dagger = \sigma_z, \quad \sigma_z^2 = I,$$

we evaluate

$$\sigma_z H(k) \sigma_z = \sigma_z [d_x(k)\sigma_x + d_y(k)\sigma_y] \sigma_z \quad (17)$$

$$= d_x(k)\sigma_z\sigma_x\sigma_z + d_y(k)\sigma_z\sigma_y\sigma_z. \quad (18)$$

Using the anticommutation relations of the Pauli matrices,

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I,$$

one finds

$$\sigma_z \sigma_x \sigma_z = -\sigma_x, \quad \sigma_z \sigma_y \sigma_z = -\sigma_y.$$

then we obtain

$$\sigma_z H(k) \sigma_z = -d_x(k) \sigma_x - d_y(k) \sigma_y \quad (19)$$

$$= -H(k). \quad (20)$$

Hence,

$$\sigma_z H(k) \sigma_z^\dagger = -H(k), \quad (21)$$

which is precisely the defining condition for chiral symmetry. Equivalently,

$$\{\sigma_z, H(k)\} = 0. \quad (22)$$

The above result is possible because the SSH Hamiltonian contains only the σ_x and σ_y Pauli matrices,

$$H(k) = d_x(k) \sigma_x + d_y(k) \sigma_y,$$

with no σ_z component, i.e.,

$$d_z(k) = 0.$$

Consequently, the vector

$$\mathbf{d}(k) = (d_x(k), d_y(k), 0)$$

is constrained to lie entirely in the d_x – d_y plane. This geometric constraint is essential for defining the winding number, which classifies the topological phases of the SSH model. Physically, the absence of a σ_z term reflects the bipartite nature of the lattice: hopping occurs only between different sublattices, while no terms couple sites within the same sublattice or produce an on-site energy difference between them.

3. Chiral symmetry is robust under variations of the hopping amplitudes, provided that the bipartite nature of the lattice is preserved. In practice, this means that the intracell and intercell hopping amplitudes may vary from bond to bond due to disorder, while the Hamiltonian continues to satisfy

$$\hat{\Gamma} \hat{H} \hat{\Gamma}^\dagger = -\hat{H},$$

as long as hopping occurs only between opposite sublattices. The chiral symmetry operator itself remains independent of these local variations in the hopping parameters.

The localization of zero-energy states on a single sublattice plays a fundamental role in the SSH model. In the topological phase, these zero-energy states appear as edge states localized at opposite ends of a finite chain. Their existence is protected by chiral symmetry, and they remain pinned at zero energy as long as the symmetry and the bulk energy gap are preserved.

2.8.1 Why Chiral Symmetry Implies an Off-Diagonal Hamiltonian

In the previous section, the SSH Hamiltonian was shown to satisfy the chiral symmetry condition

$$\{\Gamma, H\} = 0.$$

Throughout the literature, Hamiltonians possessing chiral symmetry are commonly written in the block off-diagonal form

$$H = \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}.$$

This form is not an arbitrary choice of notation. Rather, it follows directly from the algebraic properties of the chiral symmetry operator. In this section, we show that every Hermitian Hamiltonian possessing chiral symmetry can always be represented in this form by choosing the eigenbasis of the chiral operator.

Spectral properties of the chiral operator :

The chiral operator satisfies

$$\Gamma^\dagger = \Gamma,$$

and

$$\Gamma^2 = I$$

Since Γ is Hermitian, the Spectral Theorem guarantees that every Hermitian operator can be diagonalized by a unitary transformation. Consequently, there exists a unitary matrix U such that,

$$U^\dagger \Gamma U = D,$$

where D is diagonal matrix containing the eigenvalues of Γ .

The columns of U are orthonormal eigenvectors of Γ . Expressing all vectors and operators in this basis is referred to as choosing the eigenbasis of the chiral operator. Since a change of basis does not alter the physical content of the theory, we are always free to work in this representation.

What does the diagonal matrix D look like? Suppose

$$\Gamma|u\rangle = \lambda|u\rangle.$$

Applying Γ once more,

$$\Gamma^2|u\rangle = \lambda^2|u\rangle$$

Since

$$\Gamma^2 = I$$

it follows that

$$\lambda^2 = 1$$

Hence

$$\lambda = \pm 1.$$

Therefore

$$D = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

where the basis has been ordered such that all eigenvectors with eigenvalue $+1$ are placed before those with eigenvalue -1 . The identity matrices I denote the identity operator acting within the corresponding eigenspaces.

For the SSH model, the internal Hilbert space is two-dimensional, with one eigenvector corresponding to each eigenvalue. Consequently,

$$D = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus, the familiar representation of the chiral operator in the SSH model is simply the two-dimensional realization of the general diagonal form derived above.

Having established the form of the chiral operator in its eigenbasis, we now consider the most general Hermitian Hamiltonian acting on the same Hilbert space. Since the Hilbert space has been decomposed into two orthogonal eigenspaces of Γ , every linear operator acting on this space admits a corresponding block-matrix representation. In this basis, the Hamiltonian may be written as

$$H = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where the matrices A and D describe transitions within the positive- and negative-chirality subspaces, respectively, while B and C describe transitions between the two subspaces.

The defining property of chiral symmetry is that the Hamiltonian anticommutes with the chiral operator,

$$\{\Gamma, H\} = 0.$$

Using the diagonal representation

$$\Gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix},$$

we first evaluate the product ΓH ,

$$\Gamma H = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ -C & -D \end{pmatrix}.$$

Similarly,

$$H\Gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} A & -B \\ C & -D \end{pmatrix}.$$

Adding the two matrices gives

$$\Gamma H + H\Gamma = \begin{pmatrix} 2A & 0 \\ 0 & -2D \end{pmatrix}.$$

Since chiral symmetry requires

$$\Gamma H + H \Gamma = 0,$$

it immediately follows that

$$A = 0, \quad D = 0.$$

Therefore, every Hamiltonian possessing chiral symmetry must have the form

$$H = \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}.$$

Since the Hamiltonian represents a physical observable, it must also be Hermitian,

$$H^\dagger = H.$$

Taking the Hermitian conjugate of the above matrix,

$$H^\dagger = \begin{pmatrix} 0 & C^\dagger \\ B^\dagger & 0 \end{pmatrix}.$$

Comparing this expression with

$$H = \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix},$$

gives

$$C = B^\dagger.$$

Defining

$$q \equiv B,$$

the Hamiltonian finally assumes the form

$$H = \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}.$$

This result is equivalent to the projection operator formulation discussed in the following subsection.

Hence, the block off-diagonal structure of the Hamiltonian is not an additional assumption but follows directly from the existence of chiral symmetry together with the Hermiticity of the Hamiltonian.

Physical Interpretation The matrices A and D describe transitions between states having the same chirality, while B and C describe transitions between states of opposite chirality. The anticommutation relation forbids the former, forcing the diagonal blocks of the Hamiltonian to vanish. Consequently, a chiral-symmetric Hamiltonian only couples states belonging to opposite chiral sectors.

For the SSH model, these two chiral sectors correspond to the A and B sublattices. Since hopping occurs only between opposite sublattices and never within the same sublattice, the Hamiltonian naturally possesses the block off-diagonal structure derived above.

Remark The proof presented above is completely general and does not rely on any specific property of the SSH model. Any Hermitian Hamiltonian satisfying

$$\{\Gamma, H\} = 0$$

can always be transformed into the block off-diagonal form by working in the eigenbasis of the chiral operator. The SSH Hamiltonian is therefore a particular realization of this general mathematical result.

2.8.2 Consequences of Chiral Symmetry

The anticommutation relation between the Hamiltonian and the chiral operator,

$$\{\Gamma, H\} = 0,$$

imposes several important constraints on the spectrum and eigenstates of the SSH model. These consequences are discussed below.

Sublattice Symmetry An alternative and often useful characterization of chiral symmetry is obtained through projection operators onto the two sublattices. Defining

$$\hat{P}_A = \frac{1}{2}(I + \hat{\Gamma}), \quad \hat{P}_B = \frac{1}{2}(I - \hat{\Gamma}), \quad (23)$$

the operators satisfy

$$\hat{P}_A + \hat{P}_B = I, \quad \hat{P}_A \hat{P}_B = 0,$$

and project an arbitrary state onto the A - and B -sublattice subspaces, respectively.

Using the anticommutation relation

$$\{\hat{\Gamma}, \hat{H}\} = 0,$$

one finds

$$\hat{P}_A \hat{H} \hat{P}_A = 0, \quad \hat{P}_B \hat{H} \hat{P}_B = 0.$$

Thus, the Hamiltonian possesses no matrix elements connecting sites within the same sublattice. Equivalently, the only non-vanishing matrix elements are those between opposite sublattices,

$$\hat{H} = \hat{P}_A \hat{H} \hat{P}_B + \hat{P}_B \hat{H} \hat{P}_A.$$

This projection-operator formulation is completely equivalent to the block off-diagonal representation derived in the previous subsection and provides a basis-independent description of sublattice (chiral) symmetry.

Symmetric Energy Spectrum Suppose that $|\psi_n\rangle$ is an eigenstate of the Hamiltonian,

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle. \quad (24)$$

Multiplying from the left by the chiral operator gives

$$\hat{\Gamma}\hat{H}|\psi_n\rangle = E_n\hat{\Gamma}|\psi_n\rangle.$$

Using the anticommutation relation,

$$\hat{\Gamma}\hat{H} = -\hat{H}\hat{\Gamma},$$

we obtain

$$-\hat{H}\hat{\Gamma}|\psi_n\rangle = E_n\hat{\Gamma}|\psi_n\rangle,$$

or equivalently,

$$\hat{H}(\hat{\Gamma}|\psi_n\rangle) = -E_n(\hat{\Gamma}|\psi_n\rangle). \quad (25)$$

Hence, if E_n is an eigenvalue of the Hamiltonian, then $-E_n$ is also an eigenvalue. Chiral symmetry therefore guarantees that the energy spectrum is symmetric about zero energy.

Equal Support on Both Sublattices Consider an eigenstate with non-zero energy,

$$|\psi_n\rangle = \begin{pmatrix} a \\ b \end{pmatrix},$$

where a and b denote the amplitudes on the two sublattices.

Since the chiral operator is represented by

$$\hat{\Gamma} = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we obtain

$$\hat{\Gamma}|\psi_n\rangle = \begin{pmatrix} a \\ -b \end{pmatrix}.$$

Because eigenstates corresponding to different eigenvalues are orthogonal,

$$\langle\psi_n|\hat{\Gamma}|\psi_n\rangle = 0.$$

Substituting the above expression,

$$\begin{aligned} 0 &= (a^*, b^*) \begin{pmatrix} a \\ -b \end{pmatrix} \\ &= |a|^2 - |b|^2. \end{aligned}$$

Therefore,

$$|a|^2 = |b|^2.$$

Thus, every non-zero-energy eigenstate has equal probability on the two sublattices.

Zero-Energy Eigenstates The above argument applies only to states with non-zero energy. For a zero-energy eigenstate,

$$\hat{H}|\psi_0\rangle = 0,$$

there is no requirement that the state possesses equal support on both sublattices. Instead, the eigenstate may be chosen to reside entirely on a single sublattice.

This property plays a fundamental role in the SSH model, as the topological edge states appearing in the non-trivial phase are zero-energy eigenstates localized on opposite ends of the chain and occupy only one of the two sublattices.

2.9 Winding Number

The energy dispersion derived in the previous section shows that the SSH model possesses two insulating phases separated by a gap-closing point at $v = w$. Although both phases are characterized by a finite bulk band gap, they cannot be distinguished solely from the energy spectrum. A quantity that remains invariant under continuous deformations of the Hamiltonian, provided the energy gap remains open, is therefore required to classify these phases. Such a quantity is known as a *topological invariant*. For the SSH model, the relevant topological invariant is the winding number. [2–4]

From Eq. (14), the momentum-space Hamiltonian can be written as

$$H(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma},$$

where

$$\mathbf{d}(k) = (d_x(k), d_y(k), 0),$$

with

$$\begin{aligned} d_x(k) &= v + w \cos k, \\ d_y(k) &= w \sin k. \end{aligned}$$

Since chiral symmetry forbids the presence of a σ_z term, the vector $\mathbf{d}(k)$ is constrained to lie entirely in the d_x - d_y plane. As the crystal momentum k varies continuously over the first Brillouin zone,

$$-\pi \leq k \leq \pi,$$

the endpoint of the vector $\mathbf{d}(k)$ traces a closed trajectory in this plane. The topology of this trajectory provides the basis for defining the winding number.

To characterize the topology of the curve, it is convenient to introduce the normalized vector

$$\hat{\mathbf{d}}(k) = \frac{\mathbf{d}(k)}{|\mathbf{d}(k)|},$$

which lies on the unit circle. The winding number counts the number of times the vector $\hat{\mathbf{d}}(k)$ winds around the origin as k traverses the Brillouin zone once. It is given by

$$v = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\hat{\mathbf{d}} \times \frac{d\hat{\mathbf{d}}}{dk} \right)_z dk. \quad (26)$$

The subscript z denotes the component perpendicular to the d_x - d_y plane. Since the trajectory is confined to this plane, the cross product points entirely along the z -direction, and the integral measures the total angular change of the vector during one complete traversal of the Brillouin zone.

An equivalent and often more convenient formulation is obtained by introducing the complex function

$$q(k) = d_x(k) - id_y(k) = v + we^{-ik}.$$

Using this definition, the bulk Hamiltonian may be written as

$$H(k) = \begin{pmatrix} 0 & q(k) \\ q^\dagger(k) & 0 \end{pmatrix},$$

and the winding number takes the compact form

$$\nu = \frac{1}{2\pi i} \int_{-\pi}^{\pi} \frac{d}{dk} \ln q(k) dk. \quad (27)$$

Eq. (27) measures the total change in the complex phase of $q(k)$ as the crystal momentum traverses the Brillouin zone. Since the phase can change only by integer multiples of 2π , the winding number is restricted to integer values and therefore represents a topological invariant.

The winding trajectories corresponding to different hopping amplitudes are shown in Fig. 2. When the intracell hopping dominates ($v > w$), the trajectory does not enclose the origin, giving

$$\nu = 0,$$

which corresponds to the trivial insulating phase. Conversely, when the intercell hopping exceeds the intracell hopping $w > v$, the trajectory winds once around the origin, yielding

$$\nu = 1,$$

which identifies the topological phase. At the critical point $v = w$, the trajectory passes through the origin. Since the vector $\mathbf{d}(k)$ vanishes at this point, the winding number becomes ill-defined, and the bulk energy gap simultaneously closes, signalling the topological phase transition.

The winding number is a topological invariant and therefore remains unchanged under any continuous deformation of the Hamiltonian that preserves both chiral symmetry and the bulk energy gap. Consequently, two phases with different winding numbers cannot be continuously transformed into one another without first closing the gap. The gap closure at $v = w$ is therefore not merely a feature of the SSH model but a general characteristic of topological phase transitions in topological band systems [3, 4]. When the gap closes, the normalized vector $\mathbf{d}(k)$ becomes undefined at the gap-closing momentum, allowing the winding trajectory to pass through the origin and the winding number to change. After the gap reopens, the system may enter a topologically distinct phase characterized by a different winding number. This bulk topological distinction manifests itself in finite systems through the appearance of topologically protected edge states, discussed in the following section.

2.10 Bulk–Boundary Correspondence

The winding number introduced in the previous section is defined for the infinite periodic SSH chain, where translational symmetry allows the Hamiltonian to be expressed in momentum space. In contrast, the edge states discussed in the following section appear only in finite systems with open boundaries. The connection between these two seemingly different descriptions is provided by the *bulk–boundary correspondence*.

The bulk–boundary correspondence states that the existence of topologically protected boundary states is determined entirely by the topological invariant of the bulk Hamiltonian as established in the literature [2–4]. Consequently, the winding number calculated for the infinite periodic system predicts the appearance or absence of localized edge states in a finite chain.

For the SSH model, the correspondence is summarized as follows:

$$\nu = \begin{cases} 0, & \text{Trivial phase: no protected edge states,} \\ 1, & \text{Topological phase: one pair of zero-energy edge states.} \end{cases}$$

Thus, although the winding number is computed using the bulk momentum-space Hamiltonian, its physical manifestation is observed through the localized boundary states of a finite system. This remarkable connection between bulk topology and boundary physics forms one of the fundamental principles of topological condensed matter physics.

The numerical calculations presented in the following sections verify this correspondence. In the topological phase $w > v$, the finite SSH chain exhibits localized zero-energy edge states, whereas no such states appear in the trivial phase $v > w$.

2.11 Finite SSH Chain with Open Boundary Conditions

The bulk Hamiltonian derived in the previous sections assumes periodic boundary conditions, allowing the application of Bloch’s theorem and the description of the system in momentum space. While this approach is convenient for studying the bulk band structure and topological invariants, it removes the physical boundaries of the lattice and therefore cannot describe edge states. To investigate the existence of localized boundary modes, it is necessary to consider a finite SSH chain with open boundary conditions.

For a finite chain consisting of N unit cells, the periodic connection between the last and the first unit cells is removed. Consequently, the intercell hopping between the N^{th} and the first unit cell is absent, and the Hamiltonian is written as

$$\hat{H} = v \sum_{m=1}^N (|m, B\rangle\langle m, A| + \text{h.c.}) + w \sum_{m=1}^{N-1} (|m+1, A\rangle\langle m, B| + \text{h.c.}). \quad (28)$$

Unlike the bulk Hamiltonian, Eq. (28) no longer possesses translational symmetry. As a result, Bloch’s theorem cannot be applied, and the Hamiltonian must be represented directly as a finite matrix in the localized basis

$$\{|1, A\rangle, |1, B\rangle, |2, A\rangle, |2, B\rangle, \dots, |N, A\rangle, |N, B\rangle\}.$$

For a chain containing N unit cells, the Hamiltonian is therefore represented by a $2N \times 2N$ Hermitian matrix of the form

$$H = \begin{pmatrix} 0 & v & 0 & 0 & \cdots & 0 \\ v & 0 & w & 0 & \cdots & 0 \\ 0 & w & 0 & v & \cdots & 0 \\ 0 & 0 & v & 0 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & w \\ 0 & 0 & 0 & \cdots & w & 0 \end{pmatrix}, \quad (29)$$

where the alternating intracell and intercell hopping amplitudes appear as the off-diagonal matrix elements. Since the chain has open boundaries, the corner matrix elements are zero, distinguishing it from the periodic Hamiltonian.

The eigenvalues and eigenvectors of the finite Hamiltonian are obtained by solving the time-independent Schrödinger equation

$$H|\psi_n\rangle = E_n|\psi_n\rangle, \quad (30)$$

where E_n denotes the energy eigenvalues and $|\psi_n\rangle$ the corresponding eigenvectors. Owing to the finite size of the Hamiltonian matrix, the spectrum is obtained numerically by direct diagonalization. In the present work, the Hamiltonian was diagonalized using the `scipy.linalg.eigh()` routine, which computes the complete set of eigenvalues and orthonormal eigenvectors of a Hermitian matrix.

The numerical calculations were carried out for finite chains containing $N = 10$ and $N = 20$ unit cells, considering different values of the intracell and intercell hopping amplitudes. These calculations enable a direct comparison between the trivial phase ($v > w$) and the topological phase ($w > v$), and illustrate the emergence of localized edge states in the latter case.

The energy spectra obtained from the numerical diagonalization are presented in the following subsection. Subsequently, the corresponding eigenvectors are analysed to demonstrate the localization of the zero-energy states at the boundaries of the finite chain.

2.11.1 Numerical Diagonalization

The finite SSH Hamiltonian was constructed numerically for chains containing $N = 10$ and $N = 20$ unit cells. Owing to the finite dimension of the Hamiltonian matrix, the complete energy spectrum and the corresponding eigenvectors were obtained by direct diagonalization using the `scipy.linalg.eigh()` routine, which computes the eigenvalues and orthonormal eigenvectors of a Hermitian matrix.

To investigate the topological phase transition, calculations were performed for four different hopping configurations,

$$(v, w) = (0.2, 1.0), (0.5, 1.0), (1.0, 1.0), (1.5, 1.0),$$

representing the topological phase ($v < w$), a point closer to the phase transition, the critical point ($v = w$), and the trivial insulating phase ($v > w$), respectively. For each parameter set, the complete energy spectrum was calculated and the eigenstate corresponding to the lowest positive-energy eigenvalue was analysed to investigate the spatial localization of the edge states.

2.11.2 Energy Spectrum

The energy spectra obtained from the numerical diagonalization for finite chains containing $N = 10$ and $N = 20$ unit cells are shown in Fig. 3 and Fig. 4. Each point represents an eigenvalue of the finite SSH Hamiltonian.

For $v < w$, two eigenvalues appear very close to zero energy, corresponding to the topological edge states localized at opposite ends of the chain. As the hopping amplitudes approach the critical point $v = w$, the bulk energy gap decreases continuously and closes at the transition. When $v > w$, the bulk gap reopens and the near-zero-energy states disappear, indicating that the system has entered the trivial insulating phase.

Increasing the system size from $N = 10$ to $N = 20$ reduces the small energy splitting between the two edge states. This behaviour arises from the exponentially decreasing overlap between the states localized at opposite boundaries of the chain, causing their energies to approach zero with increasing chain length.

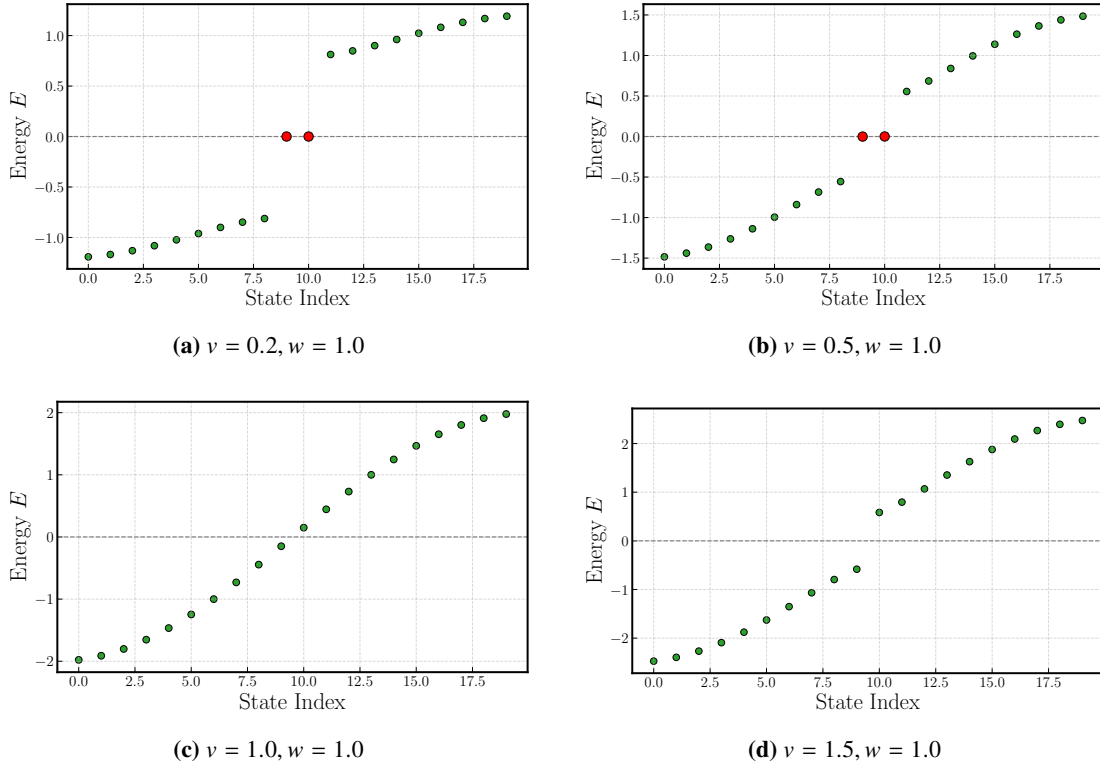


Figure 3: Energy spectrum of a finite SSH chain with $N = 10$ unit cells for (a) $v = 0.2, w = 1.0$, (b) $v = 0.5, w = 1.0$, (c) $v = 1.0, w = 1.0$, and (d) $v = 1.5, w = 1.0$. Green markers denote bulk eigenstates, while red markers indicate the two edge states. In the topological phase ($v < w$), the edge states appear within the bulk band gap and are nearly degenerate at zero energy. As the system approaches the phase transition ($v = w$), the band gap closes, whereas in the trivial phase ($v > w$), the mid-gap edge states disappear.

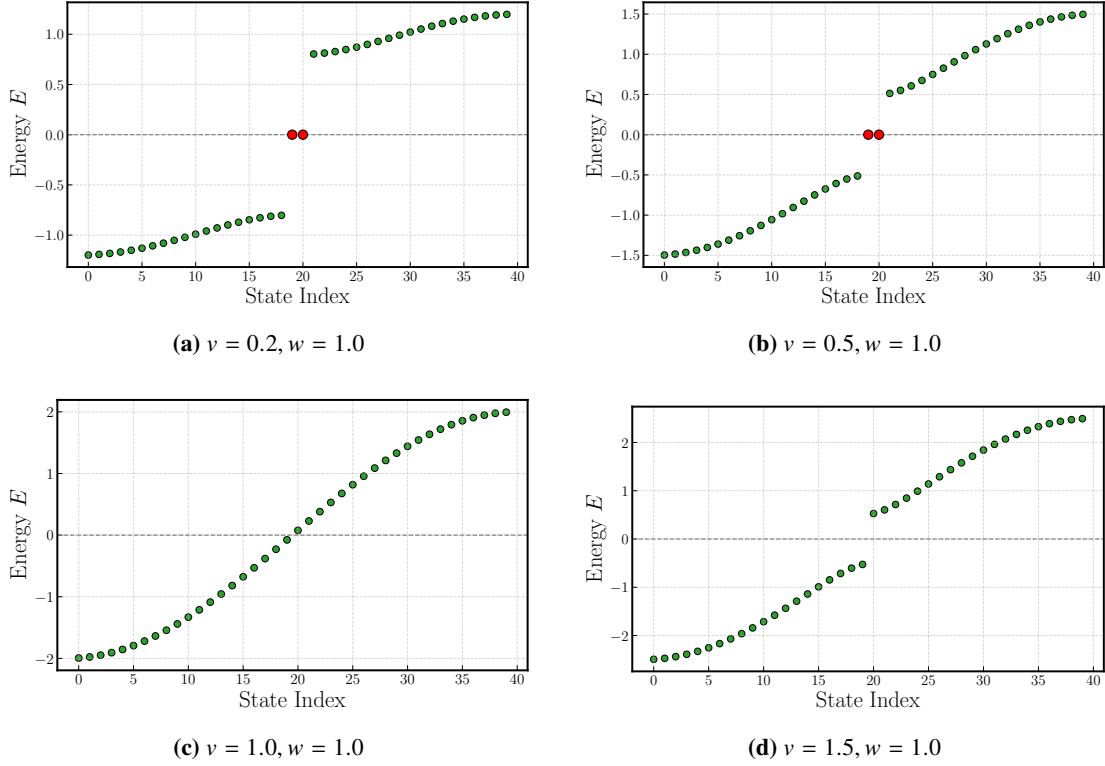


Figure 4: Energy spectrum of a finite SSH chain with $N = 20$ unit cells for (a) $v = 0.2$, $w = 1.0$, (b) $v = 0.5$, $w = 1.0$, (c) $v = 1.0$, $w = 1.0$, and (d) $v = 1.5$, $w = 1.0$. Green markers denote bulk eigenstates, while red markers indicate the two edge states. Increasing the chain length suppresses the overlap between edge states localized at opposite boundaries, resulting in a smaller energy splitting and bringing the edge-state energies closer to zero. The bulk spectrum remains essentially unchanged.

2.11.3 Edge-State Probability Distribution

To identify the spatial localization of the topological edge states, the probability density associated with the eigenstate corresponding to the lowest positive-energy eigenvalue was calculated. For each unit cell, the probabilities on the two sublattices were obtained from the corresponding eigenvector according to

$$P_A(m) = |\psi_A(m)|^2, \quad P_B(m) = |\psi_B(m)|^2, \quad (31)$$

where m labels the unit cell.

The resulting probability distributions are shown in Fig. 5. In the topological phase ($v < w$), the probability density is strongly localized near the boundaries of the chain, confirming the existence of topological edge states. As the system approaches the critical point ($v = w$), the localization length increases and the edge states extend further into the bulk. In contrast, for the trivial phase ($v > w$), the probability density is distributed throughout the lattice, indicating the absence of localized edge modes.

A comparison between $N = 10$ and $N = 20$ further shows that increasing the system size enhances the localization of the edge states by reducing their overlap, consistent with the behaviour observed in the energy spectra.

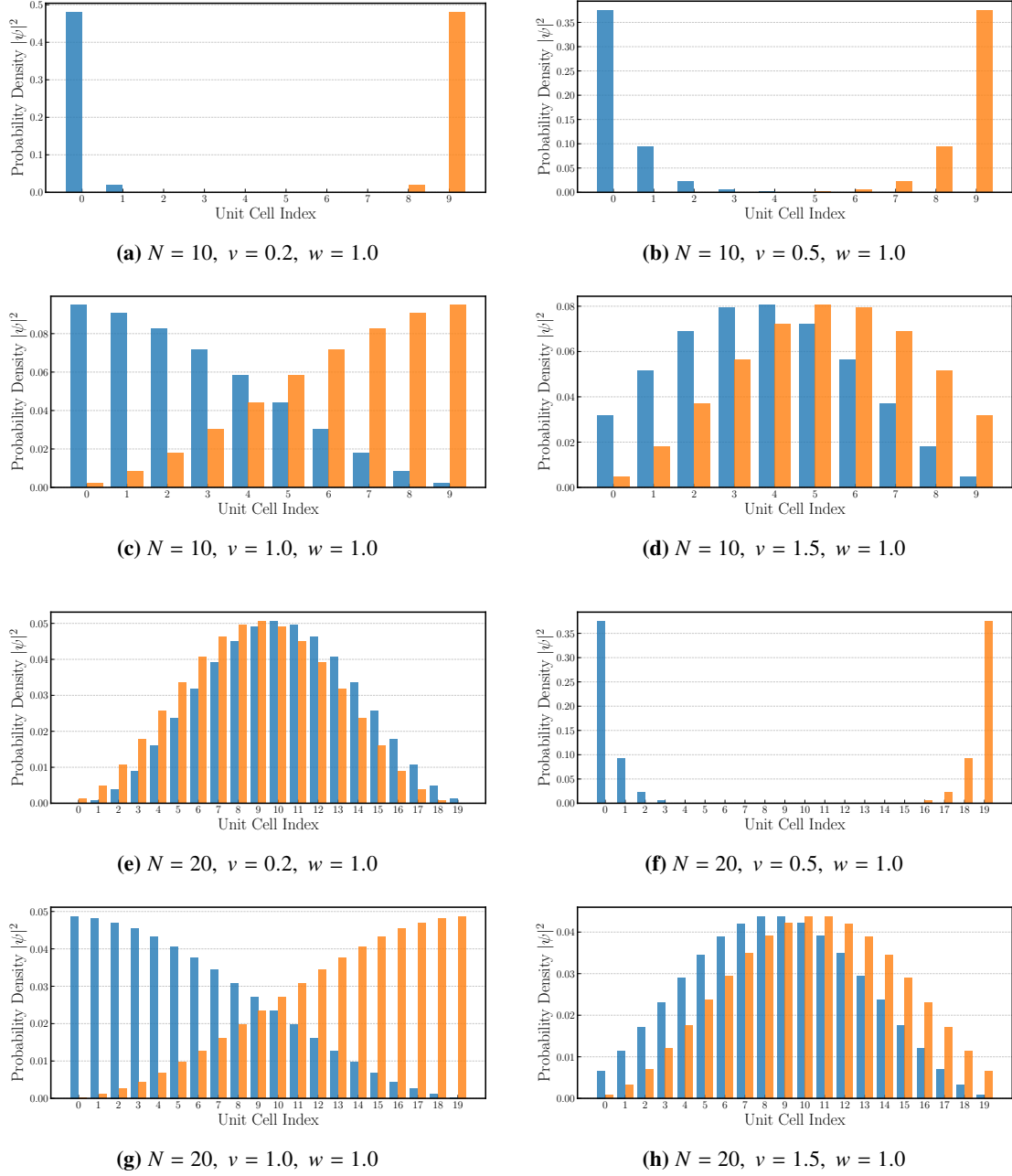


Figure 5: Probability density $|\psi|^2$ of the eigenstate corresponding to the lowest positive-energy eigenvalue for finite SSH chains containing $N = 10$ (a–d) and $N = 20$ (e–h) unit cells. The hopping parameters are $\nu = 0.2, 0.5, 1.0$, and 1.5 , with $w = 1.0$. Blue and orange bars represent the probability densities on the A and B sublattices, respectively. In the topological phase ($\nu < w$), the eigenstate is strongly localized near the boundaries of the chain. As the system approaches the critical point ($\nu = w$), the localization length increases and the state extends into the bulk, while in the trivial phase ($\nu > w$), the probability density becomes delocalized. Increasing the chain length from $N = 10$ to $N = 20$ enhances the localization of the edge states by reducing the overlap between states localized at opposite boundaries.

2.11.4 Preservation of Chiral Symmetry

The finite SSH Hamiltonian retains the bipartite structure of the infinite lattice, containing hopping only between the two sublattices. Consequently, the chiral operator $\Gamma = \sigma_z$ continues to satisfy

$$\Gamma H \Gamma^\dagger = -H,$$

or equivalently,

$$\{\Gamma, H\} = 0.$$

Therefore, the finite Hamiltonian preserves chiral symmetry. As a consequence, the energy spectrum remains symmetric about zero energy, with every positive-energy eigenstate having a corresponding negative-energy partner. The zero-energy edge states that appear in the topological phase are protected by this symmetry and remain pinned close to zero energy as long as the bulk energy gap remains open and the bipartite structure of the lattice is preserved.

3 Extension of the SSH Model: Inclusion of an On-Site Potential

In the previous sections, the SSH model was studied under the ideal assumption that both sublattices possess zero on-site energies, so that the Hamiltonian contains only nearest-neighbour hopping terms. We now extend the model by introducing a staggered on-site potential, assigning an energy $+u$ to every A sublattice site and $-u$ to every B sublattice site. This simple extension allows us to investigate how an energy imbalance between the two sublattices modifies the electronic and topological properties of the SSH model.

3.1 Real-space Hamiltonian

Including the on-site potential, the real-space Hamiltonian becomes

$$\begin{aligned} \hat{H} = & v \sum_{m=1}^N (|m, A\rangle\langle m, B| + |m, B\rangle\langle m, A|) + w \sum_{m=1}^{N-1} (|m+1, A\rangle\langle m, B| + |m, B\rangle\langle m+1, A|) \\ & + u \sum_{m=1}^N (|m, A\rangle\langle m, A| - |m, B\rangle\langle m, B|). \end{aligned} \quad (32)$$

The first term describes the intracell hopping with amplitude v , while the second term describes the intercell hopping with amplitude w . The additional term introduces different on-site energies for the two sublattices, assigning an energy $+u$ to the A sublattice and $-u$ to the B sublattice. Unlike the hopping terms, this contribution acts locally on each lattice site and does not couple the two sublattices.

Physically, the staggered on-site potential represents an energy imbalance between the two sublattices of the crystal. Instead of the two lattice sites within each unit cell being energetically equivalent, the A sublattice is assigned an energy $+u$, while the B sublattice is assigned an energy $-u$. Such a staggered potential may arise in materials where the two atoms occupying the unit cell are chemically different, through interactions with an underlying substrate, or by the application of an external electric field that breaks the equivalence between the two sublattices.

Unlike the hopping terms, which couple neighbouring lattice sites, the on-site potential acts locally on each lattice site and modifies only its energy. Nevertheless, this additional term has

profound consequences, as it breaks the chiral symmetry responsible for the topological protection of the SSH model.

3.2 Momentum-Space Hamiltonian

Since the bulk Hamiltonian possesses translational symmetry, Bloch’s theorem can again be applied [2, 5]. The external degree of freedom is transformed from the real-space basis to the momentum-space basis through the discrete Fourier transform,

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{m=1}^N e^{ikm} |m\rangle, \quad k = \frac{2\pi n}{N}, \quad n = 0, 1, \dots, N-1. \quad (33)$$

The Bloch eigenstates are therefore written as

$$|\Psi_n(k)\rangle = |k\rangle \otimes |u_n(k)\rangle, \quad |u_n(k)\rangle = a_n(k) |A\rangle + b_n(k) |B\rangle. \quad (34)$$

Projecting the bulk Hamiltonian onto the Bloch basis gives

$$H(k) = \langle k | \hat{H}_{\text{bulk}} | k \rangle.$$

The hopping terms are identical to those obtained for the SSH model,

$$\begin{aligned} \langle k | \left(\sum_{m=1}^N |m\rangle \langle m| \right) | k \rangle &= 1, \\ \langle k | \left(\sum_{m=1}^N |m+1\rangle \langle m| \right) | k \rangle &= e^{-ik}, \\ \langle k | \left(\sum_{m=1}^N |m\rangle \langle m+1| \right) | k \rangle &= e^{ik}. \end{aligned}$$

The additional staggered on-site potential contributes

$$u \sum_{m=1}^N |m\rangle \langle m| \otimes (|A\rangle \langle A| - |B\rangle \langle B|).$$

Since

$$\langle k | \left(\sum_{m=1}^N |m\rangle \langle m| \right) | k \rangle = 1,$$

the on-site potential remains unchanged after the Fourier transformation,

$$u (|A\rangle \langle A| - |B\rangle \langle B|).$$

Substituting these results into the Hamiltonian gives

$$\begin{aligned} H(k) &= u (|A\rangle \langle A| - |B\rangle \langle B|) \\ &\quad + v (|B\rangle \langle A| + |A\rangle \langle B|) \\ &\quad + w \left(e^{-ik} |A\rangle \langle B| + e^{ik} |B\rangle \langle A| \right). \end{aligned}$$

Using

$$|A\rangle\langle A| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |B\rangle\langle B| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

together with

$$|A\rangle\langle B| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad |B\rangle\langle A| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

the momentum-space Hamiltonian becomes

$$H(k) = \begin{pmatrix} u & v + we^{-ik} \\ v + we^{ik} & -u \end{pmatrix}, \quad (35)$$

Compared to the SSH Hamiltonian, the only modification is the appearance of the diagonal on-site potential $\pm u$, while the hopping terms remain unchanged.

The Schrödinger equation in momentum space is therefore

$$H(k) |u_n(k)\rangle = E_n(k) |u_n(k)\rangle,$$

which determines the bulk energy spectrum and the corresponding eigenstates of the SSH model with a staggered on-site potential.

3.3 Representation in Terms of Pauli Matrices

The momentum-space Hamiltonian may be expressed in terms of the Pauli matrices as

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y + u\sigma_z,$$

where

$$d_x(k) = v + w \cos k,$$

and

$$d_y(k) = w \sin k.$$

The additional term proportional to σ_z originates from the staggered on-site potential and represents the energy difference between the two sublattices.

3.4 Effect on Chiral Symmetry

For the original SSH Hamiltonian, the chiral symmetry condition

$$\{\Gamma, H(k)\} = 0,$$

is satisfied, where

$$\Gamma = \sigma_z.$$

For the modified Hamiltonian,

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y + u\sigma_z,$$

the anticommutator is

$$\{\Gamma, H(k)\} = \sigma_z H(k) + H(k) \sigma_z \quad (36)$$

$$= \sigma_z (d_x \sigma_x + d_y \sigma_y + u \sigma_z) + (d_x \sigma_x + d_y \sigma_y + u \sigma_z) \sigma_z \quad (37)$$

$$= d_x (\sigma_z \sigma_x + \sigma_x \sigma_z) + d_y (\sigma_z \sigma_y + \sigma_y \sigma_z) + u (\sigma_z^2 + \sigma_z^2). \quad (38)$$

Using the identities

$$\{\sigma_z, \sigma_x\} = 0, \quad \{\sigma_z, \sigma_y\} = 0, \quad \sigma_z^2 = I,$$

we obtain

$$\{\Gamma, H(k)\} = 2uI.$$

Hence, for

$$u \neq 0,$$

the operator $\Gamma = \sigma_z$ no longer anticommutes with the Hamiltonian.

One may ask whether there exists another operator that satisfies the chiral symmetry condition. Consider the most general Hermitian unitary operator

$$\Gamma = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z,$$

where

$$n_x^2 + n_y^2 + n_z^2 = 1.$$

Using the Pauli matrix anticommutation relations,

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I,$$

the anticommutator becomes

$$\{\Gamma, H(k)\} = 2(n_x d_x(k) + n_y d_y(k) + n_z u) I.$$

For chiral symmetry to exist, this expression must vanish for every value of k , giving

$$n_x d_x(k) + n_y d_y(k) + n_z u = 0.$$

Substituting

$$d_x(k) = v + w \cos k, \quad d_y(k) = w \sin k,$$

yields

$$n_x(v + w \cos k) + n_y w \sin k + n_z u = 0.$$

Since the functions 1, $\cos k$, and $\sin k$ are linearly independent, their coefficients must vanish separately. Thus,

$$n_x = 0, \quad n_y = 0, \quad n_z u = 0.$$

For

$$u \neq 0,$$

it follows that

$$n_z = 0,$$

which contradicts the normalization condition

$$n_x^2 + n_y^2 + n_z^2 = 1.$$

Therefore, no nontrivial Hermitian unitary operator exists that anticommutes with the Hamiltonian. The staggered on-site potential completely breaks the chiral symmetry, and consequently the Hamiltonian can no longer be written in a purely off-diagonal form.

3.5 Loss of the Winding Number

For the original SSH model, the momentum-space Hamiltonian may be written as

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y,$$

so that the vector

$$\mathbf{d}(k) = (d_x(k), d_y(k), 0)$$

is confined entirely to the d_x – d_y plane. As the crystal momentum varies across the Brillouin zone, the normalized vector $\hat{\mathbf{d}}(k)$ traces a closed curve in this plane, and the winding number counts how many times this curve encircles the origin.

After introducing the staggered on-site potential, the Hamiltonian becomes

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y + u\sigma_z,$$

and the corresponding vector becomes

$$\mathbf{d}(k) = (d_x(k), d_y(k), u).$$

The trajectory is therefore no longer confined to a two-dimensional plane but instead forms a closed curve in three-dimensional space. Since the original winding number is defined only for planar trajectories, it is no longer applicable once the σ_z term is present.

Thus, the staggered on-site potential breaks chiral symmetry and removes the topological invariant that distinguishes the two phases of the SSH model.

3.6 Energy Eigenvalues

The energy eigenvalues are obtained by solving the eigenvalue equation

$$H(k)|\psi(k)\rangle = E(k)|\psi(k)\rangle,$$

or equivalently,

$$\det[H(k) - EI] = 0.$$

Using the momentum-space Hamiltonian

$$H(k) = \begin{pmatrix} u & v + we^{-ik} \\ v + we^{ik} & -u \end{pmatrix},$$

we obtain

$$\det \begin{pmatrix} u - E & v + we^{-ik} \\ v + we^{ik} & -u - E \end{pmatrix} = 0.$$

Evaluating the determinant,

$$(u - E)(-u - E) - (v + we^{-ik})(v + we^{ik}) = 0.$$

Since

$$(v + we^{-ik})(v + we^{ik}) = v^2 + w^2 + 2vw \cos k,$$

the characteristic equation becomes

$$E^2 = u^2 + v^2 + w^2 + 2vw \cos k.$$

Hence, the energy eigenvalues are

$$E_{\pm}(k) = \pm \sqrt{u^2 + v^2 + w^2 + 2vw \cos k}.$$

Compared to the original SSH model, the staggered on-site potential introduces an additional contribution of u^2 under the square root. The modified expression reduces to the standard SSH dispersion relation in the limit $u = 0$.

3.7 Energy Spectrum

The energy spectrum of the finite SSH chain was computed numerically using the analytical expression derived above,

$$E_{\pm}(k) = \pm \sqrt{u^2 + v^2 + w^2 + 2vw \cos k},$$

for the parameter values

$$v = 0.2, \quad w = 1,$$

and for three different values of the staggered on-site potential,

$$u = 0, \quad u = 0.1, \quad u = -0.1.$$

The resulting energy spectra of the finite SSH chain are shown in [Fig. 6](#).

For $u = 0$, the spectrum reduces to that of the standard SSH model discussed previously, with the edge states remaining at zero energy. Introducing a finite staggered on-site potential shifts these edge states away from zero energy, while the bulk spectrum remains qualitatively similar. Since the energy eigenvalues depend on u^2 , the spectra for $u = 0.1$ and $u = -0.1$ are identical, with the sign of the on-site potential affecting the eigenstates rather than the eigenvalues.

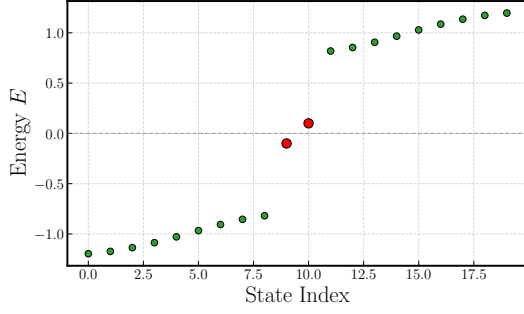
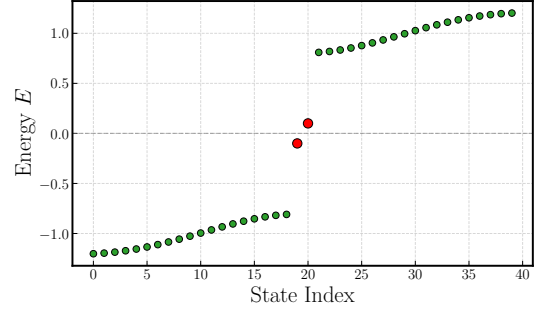
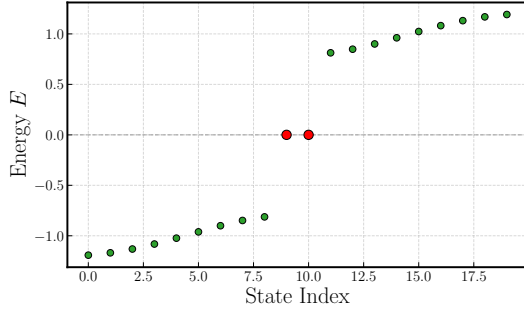
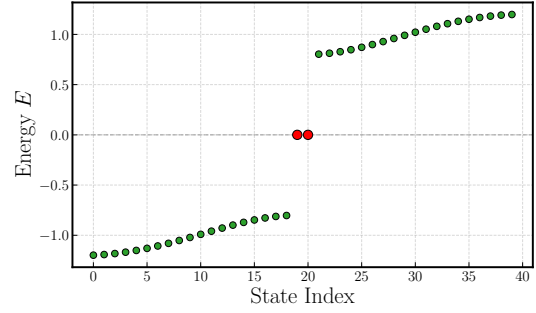
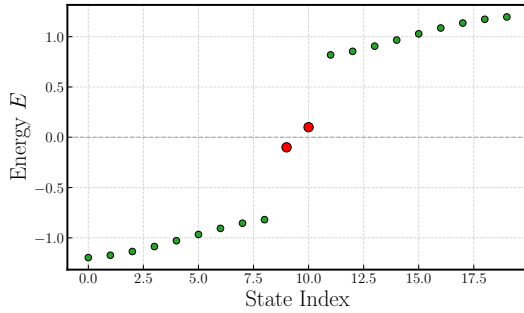
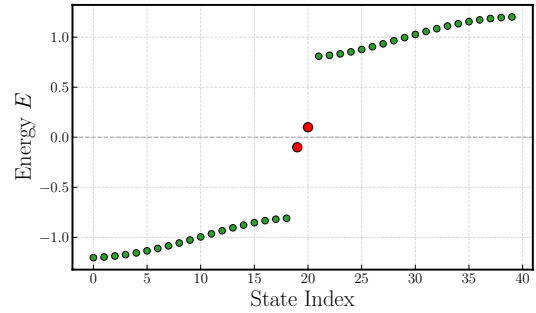
(a) $u = -0.1$, $N = 10$ (b) $u = -0.1$, $N = 20$ (c) $u = 0$, $N = 10$ (d) $u = 0$, $N = 20$ (e) $u = 0.1$, $N = 10$ (f) $u = 0.1$, $N = 20$

Figure 6: Energy spectra of the finite SSH chain with a staggered on-site potential for $v = 0.2$ and $w = 1.0$. The left column corresponds to a chain of $N = 10$ unit cells, while the right column corresponds to $N = 20$ unit cells. The rows show the spectra for $u = -0.1$, $u = 0$, and $u = 0.1$, respectively. The red points indicate the edge states.

3.8 Eigenstates

To examine the effect of the staggered on-site potential on the edge states of the SSH model, the finite-chain Hamiltonian was numerically diagonalized for

$$v = 0.2, \quad w = 1,$$

using the same values of the on-site potential,

$$u = -0.1, \quad u = 0, \quad u = 0.1.$$

The probability density associated with a normalized eigenstate is given by

$$P_n = |\psi_n|^2,$$

where ψ_n denotes the amplitude of the eigenvector at the n th lattice site. The probability distributions of the positive- and negative-energy edge states are shown in Figs. 7 and 8.

As shown in Figs. 7 and 8, the edge states remain strongly localized at the ends of the chain for all three values of the staggered on-site potential. The probability density decays rapidly into the bulk, indicating that the edge localization is largely unaffected by the introduction of the on-site potential.

For $u = 0$, corresponding to the original SSH model, the two edge states are localized at opposite ends of the chain and occupy opposite sublattices. When a finite staggered on-site potential is introduced, the edge states remain localized at the boundaries; however, the sublattice on which the positive- and negative-energy edge states are concentrated depends on the sign of u . Reversing the sign of the on-site potential exchanges the sublattice character of the two edge states while preserving their boundary localization.

Unlike the chiral-symmetric SSH model, the edge states are no longer pinned to zero energy. In the original SSH model, chiral symmetry ensures that every positive-energy eigenstate has a corresponding negative-energy partner, forcing the topological edge states to lie exactly at zero energy. The staggered on-site potential introduces a diagonal term proportional to σ_z , breaking chiral symmetry and removing the spectral symmetry about zero energy. Consequently, the edge-state energies shift away from zero while the states remain strongly localized near the boundaries for the parameter values considered.

These results show that the staggered on-site potential primarily modifies the energies and sublattice character of the edge states, while their spatial localization is largely preserved within the range of u studied.

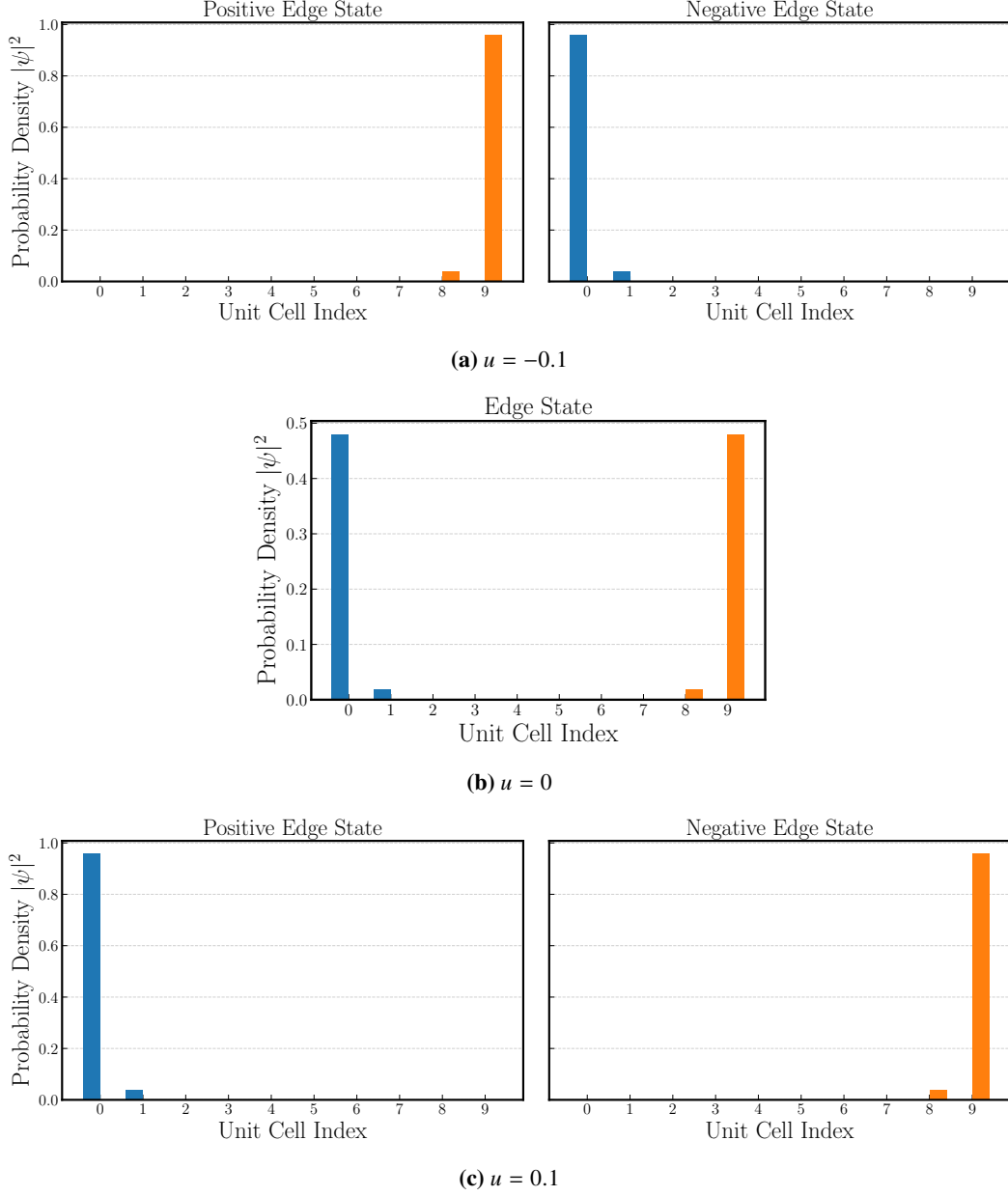


Figure 7: Probability density $|\psi|^2$ of the positive- and negative-energy edge states of the finite SSH chain containing $N = 10$ unit cells for $v = 0.2$ and $w = 1.0$. The three rows correspond to staggered on-site potentials $u = -0.1$, $u = 0$, and $u = 0.1$, respectively. Blue and orange bars denote the probability densities on the A and B sublattices, respectively. Reversing the sign of u exchanges the sublattice localization of the positive- and negative-energy edge states while preserving their spatial localization at the opposite ends of the chain.

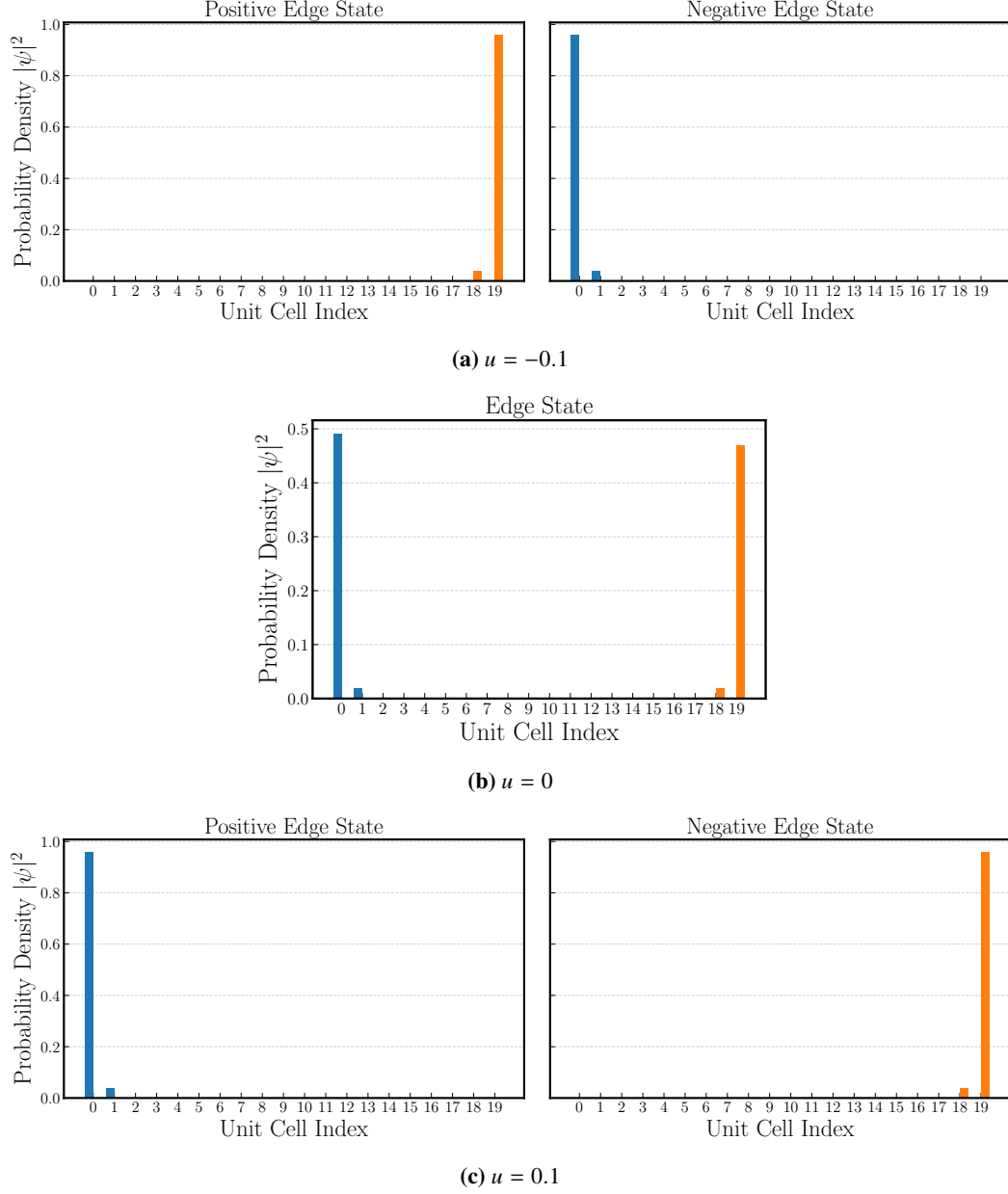


Figure 8: Probability density $|\psi|^2$ of the positive- and negative-energy edge states of the finite SSH chain containing $N = 20$ unit cells for $v = 0.2$ and $w = 1.0$. The three rows correspond to staggered on-site potentials $u = -0.1$, $u = 0$, and $u = 0.1$, respectively. Blue and orange bars denote the probability densities on the A and B sublattices, respectively. Reversing the sign of u exchanges the sublattice localization of the positive- and negative-energy edge states while preserving their spatial localization at the opposite ends of the chain.

3.9 Winding Number

For the chiral-symmetric SSH model ($u = 0$), the topological phase is characterized by the winding number, which counts the number of times the normalized vector

$$\hat{\mathbf{d}}(k) = \frac{\mathbf{d}(k)}{|\mathbf{d}(k)|}$$

winds around the origin as the crystal momentum traverses the first Brillouin zone. The winding number is given by

$$\nu = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\hat{\mathbf{d}} \times \frac{d\hat{\mathbf{d}}}{dk} \right)_z dk. \quad (39)$$

For the parameter values

$$\nu = 0.2, \quad w = 1,$$

the numerical evaluation of Eq. (39) gives

$$\nu = 1,$$

confirming that the system is in the topological phase.

When a staggered on-site potential is introduced, the momentum-space Hamiltonian becomes

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y + u\sigma_z,$$

with

$$\mathbf{d}(k) = (v + w \cos k, w \sin k, u).$$

The additional $u\sigma_z$ term introduces a finite d_z -component, so that the vector $\mathbf{d}(k)$ is no longer confined to the d_x - d_y plane. Consequently, the winding number defined in Eq. (39) is no longer well defined, since it relies on the planar winding of $\hat{\mathbf{d}}(k)$ about the origin. Instead, the effect of the staggered on-site potential is most naturally understood through the three-dimensional trajectory of $\mathbf{d}(k)$.

3.10 Three-Dimensional Winding Trajectory

For the original SSH model ($u = 0$), the vector $\mathbf{d}(k)$ is confined to the d_x - d_y plane, where its closed trajectory winds around the origin and defines the SSH winding number.

When a finite staggered on-site potential is introduced, the additional term $u\sigma_z$ generates a constant $d_z = u$ component. As a result, the trajectory is displaced uniformly above ($u > 0$) or below ($u < 0$) the d_x - d_y plane, while its projection onto the plane remains unchanged. The corresponding trajectories for $u = -0.1$, $u = 0$, and $u = 0.1$ are shown in Fig. 9.

For the chiral-symmetric SSH model ($u = 0$), the vector $\mathbf{d}(k)$ can be completely described by the polar angle

$$\phi(k) = \tan^{-1} \left(\frac{d_y(k)}{d_x(k)} \right),$$

whose total change over the Brillouin zone determines the winding number. The presence of the $u\sigma_z$ term removes this planar constraint by introducing a finite d_z -component. Consequently,

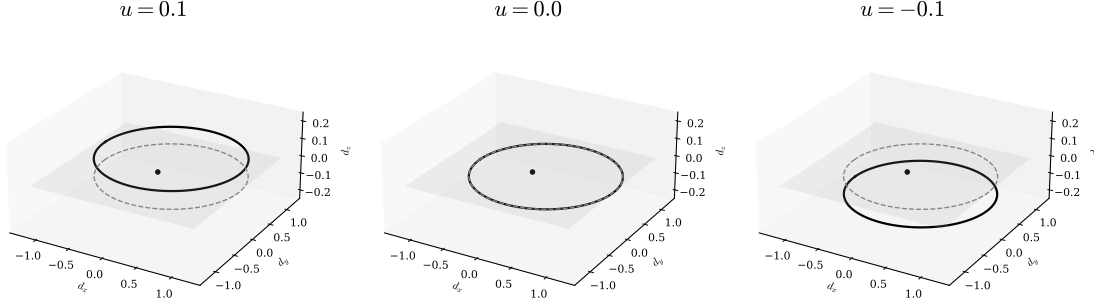


Figure 9: Three-dimensional trajectories of the vector $\mathbf{d}(k)$ for the SSH model with a staggered on-site potential for $u = -0.1$, $u = 0$, and $u = 0.1$. For $u = 0$, the trajectory lies in the d_x – d_y plane and is characterized by the winding number defined in Eq. (39). Introducing a finite on-site potential displaces the trajectory out of the plane by a constant amount $d_z = u$.

$\mathbf{d}(k)$ can no longer be parametrized by a single angular variable, and the planar winding number is no longer well defined. The three-dimensional trajectory therefore provides a more appropriate geometric representation of the effect of the staggered on-site potential and the breaking of chiral symmetry.

Table 1: Comparison of the topological and spectral properties of the chiral-symmetric SSH model and the SSH model with a staggered on-site potential.

Property	SSH Model	SSH + On-site Potential
Chiral symmetry	Preserved	Broken
Hamiltonian form	Purely off-diagonal	Contains diagonal $u\sigma_z$ term
Energy Spectrum	Symmetric about $E = 0$	Generally not symmetric about $E = 0$
d_z component	0	u
$\mathbf{d}(k)$ trajectory	Planar	Three-dimensional
Winding number	Well-defined	Undefined
Edge-state energy	Zero energy	Shifted away from zero
Edge-state localization	Localized	Remain localized

4 Conclusion

In this work, the Su–Schrieffer–Heeger (SSH) model was comprehensively investigated using a tight-binding framework to elucidate the fundamental interplay between symmetry and topology in one-dimensional electronic systems. Starting from the real-space tight-binding Hamiltonian, we derived the bulk momentum-space representation and mapped it onto the Pauli matrix basis. This vector formulation allowed for an analytical description of the bulk energy dispersion and the determination of the bulk band gap, which continuously closes and reopens at the critical transition point $v = w$. By calculating the topological invariant—the winding number ν —we successfully classified the system into a topologically trivial phase ($\nu = 0$ for $v > w$) and a non-trivial phase ($\nu = 1$ for $w > v$).

To verify the physical implications of this bulk classification, a finite SSH chain with open boundary conditions was modeled and analyzed via exact numerical diagonalization. In perfect agreement with the bulk–boundary correspondence, the topological phase manifested a pair of mid-gap states localized strongly at the boundaries of the chain. These localized edge states were shown to be a direct consequence of the internal chiral (sublattice) symmetry of the bipartite lattice, which constrains the bulk Hamiltonian to a block off-diagonal structure and strictly pins the edge-state eigenvalues to zero energy.

Furthermore, we extended the standard SSH model by introducing a staggered on-site potential $\pm u$ to explore the stability of these topological properties. Numerical and analytical treatments revealed that the on-site potential introduces a diagonal $u\sigma_z$ term that completely breaks chiral symmetry, displacing the vector $\mathbf{d}(k)$ out of the two-dimensional plane into a three-dimensional trajectory and rendering the planar winding number ill-defined. Consequently, the edge states lose their spectral protection and shift away from zero energy. Remarkably, however, their spatial boundary localization remains largely intact for small potential strengths, showcasing a nuanced distinction between spectral symmetry protection and spatial confinement.

In summary, this report demonstrates that the SSH model serves as an elegant, foundational platform for understanding topological insulators and phase transitions. On a personal note, this summer research internship has provided an invaluable opportunity to bridge the gap between abstract mathematical concepts—such as topology, chiral symmetry, and Hilbert space decompositions—and concrete physical phenomena. Through the rigorous process of analytical derivations, Python-based numerical modeling, and the synthesis of this $\text{\LaTeX}$ manuscript, this project has significantly strengthened my theoretical intuition, computational skill set, and scientific communication capabilities, laying a solid foundation for my future research pursuits in theoretical and computational physics.

Acknowledgements

I would like to express my sincere gratitude to Dr. Panch Ram, Department of Physics, Indian Institute of Technology (BHU) Varanasi, for giving me the opportunity to undertake this summer research internship and for his invaluable guidance throughout its course. I am deeply grateful for the many insightful and stimulating discussions, his constant encouragement, and his patience in addressing my questions. His mentorship not only helped me develop a strong understanding of the Su–Schrieffer–Heeger (SSH) model and its topological properties but also taught me the importance of approaching physics through careful mathematical reasoning and physical intuition.

I am particularly thankful for the emphasis he placed on understanding the underlying concepts rather than merely obtaining results. The freedom to explore ideas beyond the immediate scope of the project, together with his thoughtful suggestions and constructive feedback, encouraged me to ask deeper questions, develop greater mathematical rigor, and appreciate the fundamental role of symmetry and topology in modern physics. This internship has been an immensely rewarding learning experience that has strengthened my problem-solving skills, deepened my appreciation for rigorous theoretical analysis, and provided me with a strong foundation for future research.

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- [3] M. Z. Hasan and C. L. Kane, [Reviews of Modern Physics](#) **82**, 3045–3067 (2010).
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A Python Implementations

The following Python implementations were developed to reproduce all numerical results presented in this report. The programs generate the dispersion relations, winding trajectories, finite-chain spectra, edge-state probability distributions, and the modified SSH model including a staggered on-site potential.

A.1 Dispersion Relation and Winding Trajectories

This appendix contains the Python program used to generate the energy dispersion curves and the corresponding winding trajectories presented in [Fig. 2](#).

This program evaluates the analytical bulk dispersion relation discussed in [Section 2](#),

$$E(k) = \pm \sqrt{v^2 + w^2 + 2vw \cos k}, \quad (40)$$

and computes the vector

$$\mathbf{d}(k) = (v + w \cos k, w \sin k), \quad (41)$$

for different values of the intracell hopping amplitude v and intercell hopping amplitude w . The calculations were performed using the NumPy and Matplotlib libraries.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Hopping parameters
5 params = [
6     (1.0, 0.0),
7     (0.6, 0.4),
8     (0.5, 0.5),
9     (0.4, 0.6),
10    (0.0, 1.0),
11 ]
12
13 # Crystal momentum
14 k = np.linspace(-np.pi, np.pi, 1000)
15
16 def dispersion(v, w, k):
17     """SSH energy bands."""
18     return np.sqrt(v**2 + w**2 + 2*v*w*np.cos(k))
19
20 def winding(v, w, k):
21     """d-vector components."""
22     dx = v + w*np.cos(k)
23     dy = w*np.sin(k)
24     return dx, dy
25
26 fig, ax = plt.subplots(2, 5, figsize=(16,6))
27
28 for i, (v, w) in enumerate(params):
29
30     # Dispersion
31     E = dispersion(v, w, k)

```

```

32     ax[0,i].plot(k, E, color='black')
33     ax[0,i].plot(k,-E, color='black')
34     ax[0,i].set_title(rf"$v={v}, \, w={w}$")
35
36     # Winding trajectory
37     dx, dy = winding(v, w, k)
38     ax[1,i].plot(dx, dy, color='black')
39     ax[1,i].scatter(0,0,color='black',s=20)
40     ax[1,i].set_aspect('equal')
41
42 plt.tight_layout()
43 plt.savefig("ssh_bands_winding.pdf")
44 plt.show()

```

Listing 1: Python implementation used to generate the SSH band structure and winding trajectories.

The program evaluates the analytical expressions derived in [Section 2](#) and generates the dispersion curves together with the corresponding winding trajectories for different ratios of the hopping amplitudes. These plots are used to illustrate the closing and reopening of the bulk band gap and the change in the winding behaviour across the topological phase transition.

A.2 Finite SSH Chain: Numerical Diagonalization

This appendix contains the Python program used to construct the finite SSH Hamiltonian with open boundary conditions, perform its numerical diagonalization, and generate the energy spectra and edge-state probability distributions presented in [Section 2.11.2](#) and [Section 2.11.3](#).

The Hamiltonian matrix is constructed for a finite chain containing N unit cells with alternating intracell and intercell hopping amplitudes. The complete set of eigenvalues and orthonormal eigenvectors is obtained using the `scipy.linalg.eigh()` routine. The probability distribution corresponding to the eigenstate closest to zero energy is then evaluated separately on the A and B sublattices.

Calculations were carried out for finite chains with $N = 10$ and $N = 20$ unit cells for several values of the hopping amplitudes in order to illustrate the transition between the trivial and topological phases.

```

1
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import scipy.linalg as la
5
6 # -----
7 # Construct the SSH Hamiltonian with open boundary conditions
8 # -----
9
10 def build_ssh_hamiltonian(v, w, N):
11     """Return the 2N x 2N SSH Hamiltonian."""
12
13     H = np.zeros((2*N, 2*N))
14
15     for m in range(N):
16
17         # Intracell hopping

```

```

18     H[2*m, 2*m+1] = v
19     H[2*m+1, 2*m] = v
20
21     # Intercell hopping
22     if m < N-1:
23         H[2*m+1, 2*m+2] = w
24         H[2*m+2, 2*m+1] = w
25
26     return H
27
28
29 # -----
30 # Numerical diagonalization
31 # -----
32
33 def diagonalize(H):
34     """Compute eigenvalues and eigenvectors."""
35
36     eigvals, eigvecs = la.eigh(H)
37
38     # Find the smallest positive eigenvalue
39     positive_indices = np.where(eigvals > 1e-12)[0]
40
41     if len(positive_indices) == 0:
42         raise ValueError("No positive eigenvalues found.")
43
44     idx = positive_indices[0]
45     psi = eigvecs[:, idx]
46
47     # Calculate probabilities
48     prob = np.abs(psi) ** 2
49     prob_A = prob[0::2]
50     prob_B = prob[1::2]
51
52     return eigvals, prob_A, prob_B
53
54
55 # -----
56 # Energy spectrum
57 # -----
58
59 def plot_spectrum(eigvals):
60
61     plt.figure(figsize=(6,4))
62
63     plt.scatter(
64         np.arange(len(eigvals)),
65         eigvals,
66         s=35,
67         color="black"
68     )
69
70     plt.axhline(0, ls="--", color="gray")
71

```

```

72     plt.xlabel("State Index")
73     plt.ylabel(r"Energy  $E$ ")
74
75     plt.tight_layout()
76
77
78     # -----
79     # Edge-state probability distribution
80     # -----
81
82     def plot_probability(prob_A, prob_B):
83
84         cells = np.arange(len(prob_A))
85         width = 0.4
86
87         plt.figure(figsize=(7,4))
88
89         plt.bar(
90             cells-width/2,
91             prob_A,
92             width,
93             label="A"
94         )
95
96         plt.bar(
97             cells+width/2,
98             prob_B,
99             width,
100            label="B"
101        )
102
103        plt.xlabel("Unit Cell")
104        plt.ylabel(r" $|\psi|^2$ ")
105
106        plt.legend()
107
108        plt.tight_layout()
109
110
111    # -----
112    # Main program
113    # -----
114
115    parameter_sets = [
116
117        (0.2, 1.0, 10),
118        (0.5, 1.0, 10),
119        (1.0, 1.0, 10),
120        (1.5, 1.0, 10),
121
122        (0.2, 1.0, 20),
123        (0.5, 1.0, 20),
124        (1.0, 1.0, 20),
125        (1.5, 1.0, 20),

```

```

126 ]
127
128 for v, w, N in parameter_sets:
129
130     H = build_ssh_hamiltonian(v, w, N)
131
132     eigvals, prob_A, prob_B = diagonalize(H)
133
134     plot_spectrum(eigvals)
135
136     plot_probability(prob_A, prob_B)

```

Listing 2: Python implementation used to construct and diagonalize the finite SSH Hamiltonian with open boundary conditions and generate the corresponding energy spectra and edge-state probability distributions.

The program first constructs the finite SSH Hamiltonian as a $2N \times 2N$ Hermitian matrix with open boundary conditions by including alternating intracell (v) and intercell (w) hopping amplitudes. The complete eigenspectrum is then obtained through numerical diagonalization using the `scipy.linalg.eigh()` routine.

For each set of hopping parameters, the code generates two types of numerical results. The first is the complete energy spectrum of the finite chain, illustrating the emergence of near-zero-energy states in the topological phase. The second is the probability distribution associated with the eigenstate closest to zero energy, obtained from the corresponding eigenvector. By separating the probability density on the two sublattices, the calculations demonstrate the localization of the edge states at opposite ends of the chain in the topological regime. The generated figures correspond to those presented in [Section 2.11.2](#) and [Section 2.11.3](#).

A.3 Finite SSH Chain with On-Site Potential

This appendix contains the Python implementation used to construct and diagonalize the finite SSH Hamiltonian in the presence of an alternating on-site potential. The Hamiltonian is constructed for a finite chain with open boundary conditions,

$$\begin{aligned}
 \hat{H} = & v \sum_{m=1}^N (|m, A\rangle\langle m, B| + |m, B\rangle\langle m, A|) + w \sum_{m=1}^{N-1} (|m+1, A\rangle\langle m, B| + |m, B\rangle\langle m+1, A|) \\
 & + u \sum_{m=1}^N (|m, A\rangle\langle m, A| - |m, B\rangle\langle m, B|).
 \end{aligned} \tag{42}$$

where u denotes the alternating on-site potential. Numerical diagonalization is performed using the `scipy.linalg.eigh()` routine. The program generates the complete energy spectrum together with the probability distributions of the positive- and negative-energy edge states for several values of u .

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 import scipy.linalg as la
4
5 def ssh_hamiltonian(v, w, u, N):

```

```

6     H = np.zeros((2*N, 2*N))
7
8     for n in range(N):
9         A, B = 2*n, 2*n + 1
10
11         H[A, A] = u
12         H[B, B] = -u
13
14         H[A, B] = H[B, A] = v
15
16         if n < N-1:
17             H[B, A+2] = H[A+2, B] = w
18
19     return H
20
21
22 def edge_states(H):
23
24     eigvals, eigvecs = eigh(H)
25
26     pos = np.where(eigvals > 1e-30)[0][0]
27     neg = np.where(eigvals < -1e-30)[0][-1]
28
29     Ppos = np.abs(eigvecs[:, pos])**2
30     Pneg = np.abs(eigvecs[:, neg])**2
31
32     return (
33         eigvals,
34         Ppos[0::2], Ppos[1::2],
35         Pneg[0::2], Pneg[1::2],
36     )
37
38
39 def plot_probability(pA1, pB1, pA2, pB2):
40
41     cells = np.arange(len(pA1))
42     width = 0.4
43
44     fig, ax = plt.subplots(1, 2, figsize=(14,5), sharey=True)
45
46     for axis, A, B, title in zip(
47         ax,
48         [pA1, pA2],
49         [pB1, pB2],
50         ["Positive Edge State", "Negative Edge State"]
51     ):
52
53         axis.bar(cells-width/2, A, width, label="A")
54         axis.bar(cells+width/2, B, width, label="B")
55
56         axis.set_title(title)
57         axis.set_xlabel("Unit Cell")
58         axis.set_xticks(cells)
59         axis.grid(alpha=0.5)

```

```

60     ax[0].set_ylabel(r"$|\psi|^2$")
61     ax[0].legend()
62
63     plt.tight_layout()
64     plt.show()
65
66
67
68 def plot_spectrum(eigvals):
69
70     fig, ax = plt.subplots(figsize=(7,5))
71
72     idx = np.arange(len(eigvals))
73
74     pos = np.where(eigvals > 1e-30)[0][0]
75     neg = np.where(eigvals < -1e-30)[0][-1]
76
77     ax.scatter(idx, eigvals, s=40)
78     ax.scatter([neg, pos],
79               [eigvals[neg], eigvals[pos]],
80               color="red", s=70)
81
82     ax.axhline(0, ls="--", color="gray")
83
84     ax.set_xlabel("State Index")
85     ax.set_ylabel("Energy")
86     ax.grid(alpha=0.5)
87
88     plt.tight_layout()
89     plt.show()
90
91
92 parameter_sets = [
93     (0.2, 1.0, 0.0, 10),
94     (0.2, 1.0, 0.1, 10),
95     (0.2, 1.0, -0.1, 10),
96     (0.2, 1.0, 0.0, 20),
97     (0.2, 1.0, 0.1, 20),
98     (0.2, 1.0, -0.1, 20),
99 ]
100
101 for v, w, u, N in parameter_sets:
102
103     H = ssh_hamiltonian(v, w, u, N)
104
105     eigvals, pA1, pB1, pA2, pB2 = edge_states(H)
106
107     plot_probability(pA1, pB1, pA2, pB2)
108     plot_spectrum(eigvals)

```

Listing 3: Python implementation used to diagonalize the finite SSH Hamiltonian with an alternating on-site potential and obtain the corresponding energy spectra and edge-state probability distributions.

The program constructs the finite SSH Hamiltonian with alternating intracell and intercell hopping amplitudes together with opposite on-site potentials on the two sublattices. Following numerical diagonalization, the complete eigenspectrum is obtained and the positive- and negative-energy edge states are identified. Their probability densities are then evaluated separately on the A and B sublattices, illustrating that the edge states remain spatially localized despite their energies shifting away from zero as a consequence of chiral symmetry breaking.

A.4 SSH Model with On-Site Potential: Three-Dimensional Winding Trajectories

This appendix contains the Python implementation used to generate the three-dimensional winding trajectories presented in [Section 3.10](#). The program evaluates the modified SSH Hamiltonian in the presence of an alternating on-site potential u ,

$$\mathbf{d}(k) = (v + w \cos k, w \sin k, u), \quad (43)$$

where the additional $d_z = u$ component shifts the trajectory out of the d_x - d_y plane. The code produces the three-dimensional trajectory together with its projection onto the d_x - d_y plane for different values of the on-site potential. Calculations were performed using the NumPy and Matplotlib libraries.

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Model parameters
5  v, w = 0.2, 1.0
6  onsite_values = [0.1, 0.0, -0.1]
7
8  k = np.linspace(-np.pi, np.pi, 800)
9  dx = v + w*np.cos(k)
10 dy = w*np.sin(k)
11
12 plt.rcParams.update({
13     "font.family": "serif",
14     "mathtext.fontset": "cm",
15     "font.size": 12
16 })
17
18 fig = plt.figure(figsize=(18, 6))
19
20 # Plane z = 0
21 x = np.linspace(-1.2, 1.2, 2)
22 y = np.linspace(-1.2, 1.2, 2)
23 X, Y = np.meshgrid(x, y)
24 Z = np.zeros_like(X)
25
26 for i, u in enumerate(onsite_values):
27
28     ax = fig.add_subplot(1, 3, i+1, projection="3d")
29
30     ax.plot(dx, dy, np.full_like(k, u),
31             color="black", lw=2.5)
32

```

```

33     ax.plot(dx, dy, np.zeros_like(k),
34             "--", color="gray", lw=1.5)
35
36     ax.plot_surface(X, Y, Z,
37                    color="lightgray",
38                    alpha=0.15,
39                    edgecolor="none")
40
41     ax.scatter(0, 0, 0, color="black", s=25)
42
43     ax.set(
44         xlim=(-1.4, 1.4),
45         ylim=(-1.4, 1.4),
46         zlim=(-0.25, 0.25),
47         xlabel=r"$d_x$",
48         ylabel=r"$d_y$",
49         zlabel=r"$d_z$"
50     )
51
52     ax.set_box_aspect((1, 1, 0.45))
53     ax.view_init(elev=25, azimuth=-60)
54     ax.set_title(rf"$u={u}$", fontsize=22)
55     ax.grid(False)
56
57 fig.subplots_adjust(wspace=0.18)
58 plt.savefig("onsite_dvector_3D.pdf",
59           dpi=600,
60           bbox_inches="tight")
61 plt.show()

```

Listing 4: Python implementation used to generate the three-dimensional winding trajectories of the SSH model in the presence of an alternating on-site potential.

The program evaluates the modified $\mathbf{d}(k)$ vector for several values of the alternating on-site potential and illustrates how the winding trajectory is displaced along the d_z -direction. The projection of the trajectory onto the d_x - d_y plane remains unchanged, demonstrating that the additional on-site potential breaks chiral symmetry while preserving the geometrical projection associated with the original SSH model.